Special Issue

SSI

Social Science Information
2018, Vol. 57(3) 403–421

DOI: 10.1177/0539018418785485
journals.sagepub.com/home/ssi
SAGE

Octopuses in wild and domestic relationships

David Scheel
Alaska Pacific University, USA

Abstract

People commonly interact with terrestrial domestic animals, such as dogs and cats, horses, cattle and goats, and birds. Thereby individuals of different species form animal–human bonds. We are now forming relationships with ocean animals in increasingly common ways through growing human populations, advances in technology such as SCUBA, ocean mapping, underwater instrumentation and advances in aquatic animal husbandry. Octopuses and humans share quite distant evolutionary ties and yet share aspects of sensory ability and intelligence. Octopuses thereby pose interesting challenges and conundrums for understanding animal–human relationships. I consider several reasons to expect that the evolution of octopuses, and of animal cognition generally among active and visually sophisticated animals, will favour traits that support relationships between individuals. The evolutionary outcome of animals capable of forming inter-individual relationships may thus be expected in any evolving biota with organisms of this kind. This article explores the ability of ocean and terrestrial animals to relate to one another in ways that are reciprocal, if not equally balanced, and illustrates this with the examples of octopuses.

Keywords

cephalopod, cognition, human–animal relationships, marine, *Octopus cyanea*, *Octopus tetricus*, social behaviour

Résumé

Habituellement, les gens interagissent avec les animaux domestiques terrestres, tels que les chiens, les chats, les chevaux, le bétail et les oiseaux. Ainsi, des individus d'espèces différentes, espèces animales et êtres humains, interagissent entre eux. De nos jours, nous tissons des liens de plus en plus fréquents avec les espèces marines du fait de

Corresponding author:
David Scheel, Institute of Culture and the Environment, Alaska Pacific University, 4101 University Drive, Anchorage, Alaska, 99508-4625, USA.
Email: dscheel@alaskapacific.edu

la croissance de la population humaine, des avancées technologiques à l'instar des scaphandres autonomes, de la cartographie océanique, de l'appareillage subaquatique et de l'élevage aquicole.
Les pieuvres et les humains partagent des liens évolutifs assez distants, et pourtant ils partagent aussi une intelligence et des capacités sensorielles. Par conséquent, les pieuvres posent des défis et des problématiques intéressantes pour la compréhension des relations animal–être humain. J'examine dans cet article plusieurs raisons qui considèrent que l'évolution des pieuvres, comme celles de la cognition animale des espèces évoluées en général, favorisent des caractéristiques qui appuient l'idée de relations entre espèces. Les conséquences évolutives des animaux capables de constituer des relations interindividuelles peuvent par conséquent être envisagés pour tous biotes évolutifs avec des organismes de ce type. Cet article explore à travers l'exemple des pieuvres, la capacité des animaux marins et terrestres d'interagir réciproquement les uns avec les autres, mais sans pour autant que la relation soit tout à fait équilibrée.

Mots-clés
céphalopode, cognition, relations humains–animaux, maritime, *Octopus cyanea, Octopus tetricus*, comportement social

Background

People form relationships with domestic animals. The ways in which domesticated animals are valued by people are varied, and include providing labour (e.g. from horses, hunting dogs and cats controlling mice), food (e.g. from chickens, cattle, goats, pigs, farmed salmon and tilapia), as a hobby (e.g., pigeon breeders), animal-assisted therapy (e.g. with dogs and horses), and for companionship (particularly from dogs, cats, fish and birds). The process of domestication for most species involves genetic change via artificial selection, but crucially also at least a tolerance for living the entire life-cycle near people (Driscoll et al., 2009).

Opportunities for people to form individual relationships with wild animals are more limited, but this can also occur. However, taming an individual wild animal (to tolerate a human) is not the same as domestication of a species or strain.

People acknowledge similarities with animals when we recognize the intelligence and communication abilities of other large-brained and social mammals such as chimpanzees and dolphins. We have been surprised when similar abilities can be developed in parrots, which do not have notably large brains, but have many behaviours that we admire as 'smart'. However, regardless of absolute brain size, birds and mammals share a high encephalization quotient relative to other vertebrates (Jerison & Barlow, 1985). What sort of other animals might share cognitive similarities with people in interesting ways?

Here I argue somewhat speculatively that the ability to form relationships with other individuals is widespread among animals whose sensory and cognitive ability construct a complex reality. A relationship is the stance or attitude with which two or more connected beings regard each other. We should expect the ability to have relationships to evolve wherever certain types of animals do – those that are mobile and flexible interacting with their environment. This view is distinct from that of an anthropocentric *scala naturae* in which humans occupy a privileged spot at the pinnacle of nature and other animals are merely more or less like us. Instead, natural selection favours adaptations among species, according to the different pressures in their environment, such that flexible cognition will evolve where it is adaptive to the animal's particular life history (e.g., Vitti, 2013). Here, I will use octopuses as exemplar cases for the view that, with regard to relationships, these conditions occur among animals with a particular set of life-history circumstances. There are a few reasons to expect that the ability to form relationships is limited to mammals in particular, or terrestrial animals, or everywhere among the vertebrate animals, or just those animals with parental care of young.

Should it be expected that some invertebrate animals such as octopuses can form relationships with individuals? When kept in aquariums, octopuses may behave differently towards different people (Anderson et al., 2010). Octopuses can do this because they can communicate in expressive ways through skin colour, skin texture and complex behaviour such as shooting a jet of water at a disliked aquarist (e.g., Mather, 1992). That they behave this way reveals that octopuses recognize individuals around them as more than just objects in the environment. Octopuses recognize other octopuses, fish, prey, predators or aquarists as agents – other animals that, like octopuses, have their own intents (Scheel et al., 2018). Octopuses appear also to do this in the wild. This recognition can explain why an octopus will lazily reach out to grab a swimming scallop, which has limited agency in directing its escape, but will stealthily approach and then abruptly pounce on a fish, which can direct its own escape very precisely if it has time to react to a threat. To react appropriately in each case, the octopus must identify agents as members of a category – this one is slow, that one is fast and agile, another is a threat. Many octopuses for example, will approach and interact with a curious SCUBA diver in ways they do not with an investigating sea lion (for an example, see Godfrey-Smith, 2016). In this sense, octopuses recognize and work with categories or concepts.

Octopuses recognize categories of predator and react differently to each. Among the categories of predators that threaten octopuses are cursorial, ambush and mobbing predators. Cursorial predators move quickly, but may not remain in an area for long. This category includes sharks, dolphins and sea lions. Ambush predators remain stationary for long periods, and attack prey with a sudden lunge. The wobbegong shark is an example of an ambush predator. Mobbing predators may not individually be capable of subduing an octopus, but attack in numbers, which can overwhelm the octopus. In Australia, a diver filmed an octopus being mobbed and killed by a swarm of leatherjackets.[1]

At a study site in Jervis Bay, Australia, a few colleagues and I have been recording video of gloomy octopuses (*Octopus tetricus*) interacting in the wild with predators, prey and with each other (Godfrey-Smith & Lawrence, 2012; Scheel et al., 2014, 2016, 2017). Audrianna Kinley (2015) reviewed video of predators approaching these octopuses at their dens. When approached by a cursorial predator, octopuses do not react or react little. As sharks, dolphins and sea lions swam by, octopuses in Jervis Bay were momentarily stationary, and may have pulled slightly down. In contrast, when a wobbegong shark, an ambush predator, was visible nearby, these octopuses ceased activity all day, and spent the time out of sight in their dens when they otherwise would have been active.

One octopus emerged hesitantly from a den over many minutes, and retreated quickly again to shelter when the wobbegong shark approached it. Finally, an active octopus bitten by a single leatherjacket, a mobbing predator, spent more time in its den after the bite than it did before but in a few minutes resumed normal activity. To react in different ways to predator threat suggests the octopuses are categorizing the other animals, and reacting to each category of threat differently. This is a subtler distinction than that between prey, predator or mate. I suspect that similar categorization occurs within each broader category.

Curious, active, complex, sensory individuals

Octopuses are curious and active, with complex bodies and diverse senses. Curiosity may serve the purpose of obtaining enough information about the world to sort other animals into appropriate categories. The octopus uses primarily vision for this purpose (Gleadall & Shashar, 2004). Vision is a distance sense with no lag time, often providing the earliest notice of an approaching change in the environment. Vision also includes multiple channels of information such as form, colour, and motion. Octopuses and cuttlefish can separate objects from background in their worlds using mechanisms of opponency, motion and feature-detection in ways similar to mammals and birds (e.g. Gleadall & Shashar, 2004; Hanlon & Messenger, 1996; Zylinski & Osorio, 2014). These capabilities may exceed those of amphibians and some insects to recognize a variety of objects because, for example, amphibian and preying mantid abilities to recognize prey depend on specific relations of object movement relative to body geometry (Ewert, 2004; Kral & Prete, 2004), and appear to be less general than the ability of octopuses and cuttlefish to recognize objects. As an example, octopuses are also capable of visually directing arm movements (Gutnick et al., 2011).

The lateral line analogue systems of cephalopods are able to detect water movements that, for example, enable cuttlefish to catch small prey nearby in darkness. Octopuses also rely on touch and smell or taste from chemoreceptors concentrated, among other places, in the rims of their suckers. The touch–taste sensory world of an octopus allows it to find and identify suitable food such as crabs in crevices in its habitat where it cannot see, pull that food under its web and get through any protective shell to make a meal of it.

To accomplish these tasks the octopus must use its complex and flexible suckers and arms, with which it can probe the crevices, grip and pull a clam shell, walk across the reef, use its arm crown as part of a hydroplane to move through the water, groom, excavate and maintain a den, display, fight or mate. The body also includes a mantle and siphon that are used in respiration and for jet propulsion and directing a jet of water for digging or burying, removing debris and deterring unwanted approaches. The beak and radula, used during feeding, add to the suite of actions that are supported by octopus anatomy. Over the aboral surface of the body, octopuses muscularly control skin texture and colour to change appearances, camouflage or display signals, capabilities that seem to be used to avoid predators (e.g., Josef & Shashar, 2014), facilitate prey capture (Mather & Mather, 2004) and engage with conspecifics (e.g., Huffard, Caldwell & Boneka, 2008; Scheel et al., 2016).

Roles, recognition and relationships

Might octopuses take on roles in interactions with others, recognize others and form relationships with others? What evidence is there that octopuses are capable of these behaviours? There are anecdotes from captive animals, and a few scientific studies. Although careful and detailed analyses remain to be done, at the study site in Jervis Bay I have seen signs that these octopuses may be engaged in all of these.

That cephalopods adopt roles is not surprising. At its simplest, that octopuses attack food, shrink from predators, and recognize potential mates demonstrates adaptive reactions to context during interactions with another. At this level, being in a role is ubiquitous in the biological world. Octopuses also adopt more complex roles. Octopuses pay attention to fish, who pay attention to them. For example, an octopus has been observed tolerating attention from cleaning gobies (Johnson & Chase, 1982). In a more complex case, coral trout sometimes hunt collaboratively with octopuses on the Australian great barrier reef. Coral trout are large fast fish, but cannot hunt prey in the holes and crevices of the reef as can octopuses. The coral trout will shimmy their bodies in front of the den of the octopus to signal starting a joint hunt, repeating the signal until the octopus joins them. When prey is found but escapes, the coral trout may do a head stand (orienting vertically) with periodic headshakes over the new hiding place of the prey, until the octopus approaches to reach into the crevice that is out of reach of the coral trout. Octopuses are much more likely to approach coral trout after the coral trout does a headstand than in the absence of the signal (Vail et al., 2013).

Other cephalopods are known to assume roles in social interactions (beyond merely male and female roles during mating). Adamo and Hanlon (1996) report that cuttlefish signal to one another, and Hanlon et al. (2002) report on the mating system of *Loligo vulgaris* in which the squids adopt complex roles such as guarding males that prevent intruder males from gaining access to a female, and sneaker males that imitate females to gain mating opportunities. These roles are responses to an individual assessment of its own status or abilities in relation to others in the environment.

In a preliminary look at the behaviours of *Octopus tetricus* in Jervis Bay, my colleagues and I have reported on limited ways in which these octopuses also act in roles. Gloomy octopuses signal to each other, and these signals predict the outcomes of agnostic interactions (Scheel et al., 2016). Octopuses turned dark, stood tall, raised the mantle over their eyes, spread their web and arms, and displayed on high ground. These behaviours correlated with approaching another octopus and, when the other octopus also displayed some of these behaviours, an increased chance of the two animals grappling. Another common response to this signal was the other octopus crouching low and flat and displaying a high contrast deimatic body pattern, behaviours associated with the deimatic octopus avoiding further interaction or contact between the two individuals by retreat or moving towards a den.

In the video abstract of that report, we show a brief interaction among five gloomy octopuses. The central octopus occupies the stand tall/aggressive role. A second octopus comes on-screen displaying the deimatic posture and when approached, retreats into the distance. A third octopus also displays deimatic posture and retreats from the first octopus. Two additional octopuses reach out toward the first from within their dens. The fourth reaches with a dark arm and contacts the first (a brief grapple). The fifth reaches out with a pale arm (part of the deimatic display) and no contact results with the first octopus. This brief sequence suggests that occupying roles within interactions among octopuses is common at this site.

The ability of octopuses to recognize others has been discussed in the scientific literature. Boal (2006) reviewed the evidence for recognition among cephalopods and, as mentioned above, found that at least some cephalopods are capable of recognizing species, sex, sexual receptivity and possibly dominance. However, she found no evidence in support of kin or individual recognition at that time. Octopus can exhibit inter-sexual aggression (e.g. for *Octopus tetricus* Huffard & Godfrey-Smith, 2010) and sexual cannibalism (e.g., for *O. cyanea* Hanlon & Forsythe, 2008; Huffard & Bartick, 2015; Ibáñez & Keyl, 2010). As such, it is not surprising that mate guarding has been little-reported in octopuses, although for an exception see Huffard, Caldwell and Boneka (2008 for *Abdopus aculeatus*). Mate guarding may require individual recognition, but guarding may also be accomplished via other cues such as proximity, area recognition or sex recognition. Following Boal's (2006) review, Anderson et al. (2010) found that *Enteroctopus dofleini* in aquariums recognized individual caretakers, and Tricarico et al. (2011) concluded that *O. vulgaris* in captivity recognized their neighbours for at least one day. However, these are the only reports of this capability.

In our video from Jervis Bay, gloomy octopuses in neighbouring dens often tolerate each other, but commonly there is also a great deal of jostling and low-level harassment. In one bit of video, a male octopus approached a den. This den was occupied and, as the male approached, the other octopus pulled down inside the den until out of sight. The approaching male reached in, grappled with and evicted the occupant, who was another male (Scheel et al., 2017). The evicted octopus fled to another den a few meters away. After a few minutes, the evicting octopus made a short excursion away from the den sites, far enough that the individual dens were out of site. Then this male jetted back directly to the den to which the evicted male had fled, and again evicted it from its den.

Thus, in a period of about ten minutes, a male octopus arrived at a den and evicted its occupant, who retreated to another den. After an interval, the male proceeded directly to the newly occupied den and again evicted the same male. The evicting octopus made an excursion between the two evictions, and thus was not merely visually tracking the other octopus. This observation is suggestive that these octopuses recognize each other *as individuals*, more than (merely) by proximity or by the role an octopus assumes in an interaction, behaviours that are consistent with what we are seeing in much of the other video still being analysed.

The ability of octopuses to form individual relationships has little been addressed, although we might take from earlier reviews (Boal, 2006; Tricarico et al., 2014), that while there are interesting hints in a few studies, there are as yet little positive data showing this to be the case. Anecdotally, caretakers of octopuses in captivity often report that the octopuses know them and have related differently to different people (e.g., Montgomery, 2015). However, sometimes this may be dismissed as anthropocentrism, our human tendency to unjustifiably attribute human capabilities or perspectives to non-human animals or even machines (such as a car or computer). Further, there is little written about relationships among captive octopuses themselves (but for exceptions regarding

Figure 1. Octopuses in Jervis Bay.
(Credits: David Scheel, Peter Godfrey-Smith, Matthew Lawrence, Stefan Linquist).

mating, tolerance and dominance, see Boyle, 1980; Caldwell et al., 2015; Cigliano, 1993; Forsythe & Hanlon, 1988; Mather, 1980, 1985 and references mentioned above). Octopuses in captivity are normally housed separately due to the risk of cannibalism. Thus little opportunity may exist for captive animals to demonstrate relationships with conspecifics.

In a video from Jervis Bay, at least one other observation of interactions among gloomy octopuses is suggestive. In this instance, the assigned sexes are suggested only based on behaviour and some anatomical traits, but the full video has not yet been analysed in detail.

A female (Figure 1, octopus F) and male octopus (Figure 1, octopus M) were in adjacent dens. Octopus M pulled down into his den and was lost from view. In the background, a third octopus in a den a few meters away was approached by a fourth octopus, a presumed rival male entering from off screen. The octopus F left her den and grappled with this other male at the den of the third octopus. She then returned to her original den. This other male then approached and entered the den of the octopus M adjacent to octopus F. Octopus M and the other rival male (Figure 1, octopus R) grappled, and octopus

M was evicted from his den. In the tangle of octopus limbs on video, it is difficult to see which octopus was evicted, but careful attention to the eyes and the arms revealed that octopus M (lower in the den) left the den by moving past octopus R (overlying) who entered the den last. At this point (Figure 1, top panel), octopus F entered the den with octopus R and evicted him from the den. As he retreated from her, she attempted to restrain him (Figure 1, lower panel) and wrap an arm under the mantle. This may have been a failed attempt at a strangle hold (Huffard & Bartick, 2015). She returned to her den and octopus M, who had been nearby, re-entered his den after brief contact with the female. The octopus R returned to another den out of arms' reach from the original adjacent octopuses' F and M. The video does not reveal whether octopus R had previously occupied the den earlier in the sequence, but that would be consistent with other behaviours we have seen on film.

In this account, both male octopuses momentarily were in the same den. One left, but the female did not allow the remaining male to stay. Once he departed, the original male was tolerated by the female in the den adjacent to hers. From this it appears that the female recognized the male she tolerated, and may also have recognized as a rival the second male, given her earlier excursion from her den to grapple with him when this male came onscreen. Her preference for one male over the other as a neighbour (possibly a mate) suggests a relationship between this female and one or both males. Such a relationship would be possible when built on individual recognition of other octopuses. Mate selection by males and females occurs among many animals including cephalopods (Boal, 1997; Huffard et al., 2010). However, mate selection may not depend on individual recognition and can rely on indicators of mate quality (such as size, display intensity or chemical signals, e.g. Morse et al., 2017).

The aware octopus

Octopuses are the only invertebrate group named in the Cambridge Declaration of Consciousness (Low et al., 2012) as having the neurological substrates of consciousness. Recognition of agency in self and others is a hallmark (perhaps the defining hallmark) of consciousness. I took for granted that octopuses recognize agency in the above discussion of roles, recognition, and relationships. Beyond the behaviours described above, is there evidence that octopuses recognize agency, either in others or in themselves?

When verbal self-reports of consciousness (that a speaking person could give) are not available, as they are not from animals, some studies have inferred consciousness as a theory of mind when an animal can discriminate behaviourally based on what another has access to know. Thus apes will behave differently in interactions when they have seen another ape watch someone hide food than they will when the food hiding was visible only to themselves (for review, see Call & Tomasello, 2008). To achieve this discrimination, the ape must know what the other ape knows and bear in mind that the other ape will act on that information. In short, the animal must infer another mind.

The ability to infer another mind is similar to (and may depend on) recognizing agency. Dennett (2008) has referred to this as the intentional stance, to recognize agency in certain kinds of individuals by employing an internal model that predicts one sort of future episode rather than another. If the cognitive model includes awareness of oneself and ones' goals in relation to others, Dennett argues, perhaps that is constitutive of one level of consciousness. (A human-like level might further include the more nuanced awareness of another's beliefs and goals, including another's false beliefs and desires.) Vitti (2013) has argued that awareness of agency is a by-product of the evolution of flexible intelligence.

The more human-like consciousness is higher-order consciousness, and is often contrasted with a more fundamental consciousness. Being mentally aware of anything is called primary consciousness (Griffin & Speck, 2004). Primary consciousness entails that objects of perception be united in 'episodic scenes' (Edelman et al., 2005), and that in turn seems to demand a kind of internal model about the environment like Dennett envisions, as mentioned above. Do octopuses have this kind of awareness of themselves and their own interests in relation to others? The anecdotes above hint in this direction, but here I will use parallels between octopus and social mammal behaviours as the basis for the discussion. African lions are a social but non-primate mammalian example. Mammals are generally acknowledged to possess primary consciousness (Edelman et al., 2005). One theory of consciousness attributes it to complex social relationships, so lions make a good example of a mammal we understand to be conscious in this sense. Borrego (2017) offers big cats, that vary considerably in social systems as a felid model to study the role of sociality in the evolution of intelligence. If octopus species also vary in social interactivity, the octopuses could provide a similar model system among invertebrates.

Lion behaviour has been the subject of continuous study in the Serengeti National Park, Tanzania since at least 1966 (e.g. Hanby, 1982; Schaller, 1972). Since that time, the Serengeti Lion Project has kept records on every lion in the population, tracking individual lions throughout their lives. The lions live in stable social groups of female relatives and young accompanied by cooperative cohorts of related males. The researchers can recognize them individually and no doubt they recognize their pride mates as well.

Pride mates hunt together, but while doing so they do not all behave the same. I found that there were three different strategies of lions during hunts in the Serengeti (Scheel & Packer, 1991): female lions were Pursuers leading the stalk, Copycats moving in synchronicity with the other lions, or Refrainers that did not stalk the prey but still shared in a kill if the hunt was successful. Females were less likely to refrain from hunting with their pride mates when the prey was difficult, such as a large buffalo; and overall males were more likely than females to refrain. Refrainers during the stalk were also likely to refrain from the rush, when the prey was pursued and captured.

The awareness of lions during a hunt includes lion expectations of the environment, of themselves and of other lions. Refrainers were less likely when prey was difficult, therefore lions must monitor the chance that the hunt would succeed. Lions monitor their own role during a hunt. Their behaviour was consistent between stalk and capture but was not always the same from hunt to hunt for a given lion. They also monitored their spatial position relative to other hunters, otherwise Copycats would not have been possible, which also required lions to monitor the behaviour of their pride mates, not just themselves. Once one lion began to stalk an easy prey such as a warthog, the others refrained. When hunting a difficult prey such as a buffalo, Copycat hunters stayed with the groups using frequent visual checks on pride mates. Refraining while another hunts indicates the behavioural expectation that '*she* will hunt and then *I* will eat'. These behaviours are consistent with and support that expectation that as social mammals, lions possess consciousness.

How does the behaviour of an octopus compare? Octopuses are invertebrate, generally solitary, and often considered asocial: our expectation of their awareness of environment, self and others has been lower. Compared to mammals, Edelman et al. (2005) considered the octopus 'a distant case and a major challenge'. I will advance three anecdotal observations and argue that these indicate octopuses may have a similar awareness to lions of their environment, themselves and themselves in relation to others. First, as mentioned above, octopuses lunge at fish but not at less mobile prey. Second, their body patterns anticipate their actions, revealing intentions. Third, octopuses attend to the attention of others. As an example of this, octopuses will manoeuvre themselves to place an object between themselves and an observer, but then bring just their eyes back into view, indicating an awareness of their body in relation to the perspective of the observer.

Like lions that are aware of the difficulty of their prey, octopuses are responsive to the typical escape response of different prey so that they use appropriate attack behaviour. Octopuses, including *Octopus cyanea*, often forage by touch (tactile and chemoreception) among coral crevices (Forsythe & Hanlon, 1997). When doing so, they capture non-mobile prey or mobile animals hiding within the reef. Fish may hover nearby, either not attending to the octopus or anticipating the possibility of food scraps or the disturbance of other potential prey items that the fish might snatch up (e.g., Diamant & Shpigel, 1985). Sometimes an octopus will try to capture a fish, but to do so the octopus must attack differently to defeat the streamlined fish escape response. Octopuses in each case may use a behaviour called a web-over, in which the arms are spread and the web lowered over the substrate like a net. Between web-overs in normal foraging, the octopus moves gradually along the reef without pouncing. The colour of *O. cyanea* during this movement is typically dark, sometimes with a white dorsal and mantle stripe (a disruptive pattern). However, during web-over capture attempts, the web, and often arms, head and mantle, are pale (Figure 2).

When attempting to capture a fish, *O. cyanea* pales and makes an abrupt pounce. The colour change links the abrupt pounce to web-over capture attempts. The octopus is trying to capture the fish. The unique pouncing attack anticipates the fish escape response, and is an attempt to engulf the fish under the web before it has time to react. By pouncing the octopus reveals its expectations of another agent in the environment, in this case the fish. The colour change preceding the pounce suggests an awareness of their own intention to pounce. *Enteroctopus dofleini* will pounce when attempting to capture shrimp, which also attempt escape with abrupt directed movement (from personal observation).

A different body pattern by another octopus species also reveals an awareness of self within a group of fish. Off the coast of Brazil, *Octopus insularis* is often followed by several small groupers. When on the bottom, the octopus maintains a camouflage pattern typical of many foraging octopuses. However, when the octopus leaves the bottom to swim from one area to another, the octopus adopts body patterns that resemble the fishes among which it is swimming (Krajewski et al., 2009). Here we see the octopus again adopting a body pattern revealing its own intent – to forage along the bottom or briefly to swim with fishes, adopting a fish-mimicking pattern that reduces its conspicuousness

Figure 2. Blanching white.
(Credit: David Scheel).

in the water column. These octopuses also typically swim in the middle of the group of fish, rather than leading or following, a position of greater safety.

This same awareness of themselves is demonstrated again when octopuses signal their intent to stand ground or retreat during interactions with other octopuses (Scheel et al., 2016). In each of these cases, the body pattern of the octopus at one moment indicates how it intends to act next (sudden pounce at a fast prey, swimming in a school, standing ground against an approaching intruder). Like the lion whose behaviour during a stalk predicts its participation in a fast rush at the prey, these colour changes indicate the octopus has expectations of its own future behaviour.

Like lions that watch their pride mates, and adjust their behaviour accordingly, octopuses also adjust their behaviour in ways that indicate an awareness of agency in others. Although this is clear in the context of signalling (Scheel et al., 2016), as described above, it also applies to interactions with their predators (Kinley, 2015). In these cases with conspecifics and predators, and also when octopuses interact with humans, they attend to the attention of others. For example, an octopus will respond to an image of a sea lion looking at the camera as it would to an agent of concern in the environment – the octopus ceases other activity and creeps slowly into hiding (from personal observation). As a SCUBA diver, I have repeatedly observed the following sequence when encountering an octopus that is exposed and away from shelter. When the diver detects the octopus, the octopus freezes (perhaps for many seconds). The diver attends to the octopus. The diver may momentarily divert attention away from the octopus, at which cue the octopus makes an abrupt escape. The octopus breaks cover or camouflage, adopts a dark colour for open water swimming (Forsythe & Hanlon, 1997), jets away, abruptly changes direction, possibly inks and then becomes immobile on the bottom, switching to a cryptic

Figure 3. *O. cyanea* peaking
(Credit: David Scheel).

body pattern. The sequence of events when an octopus returns to its den, for example, differs from this in typical colour displayed, use of crypsis, and whether the octopus walks or freezes abruptly at the end of the swim.

If the diver continues to follow and attend to the octopus closely, the octopus often will move to place an object between itself and the diver (Figure 3).

The diver may lose sight of the octopus behind this obstruction, but, in a moment, the octopus often raises its eyes or adjusts its position until it can observe the diver. These behaviours result in the octopus hiding its body from the diver, but keeping the diver in view to observe. Thus, the octopus orients its eyes toward one of several competing stimuli in the environment – it is attending to the diver. In addition, by hiding its body, it has acted on the presumed visual perspective of the diver. This suggests awareness of itself in relation to the diver and the diver's perspective.

Sceptics may argue that the octopus follows a rule of thumb, such as 'move around a large object until the threat is no longer visible', and judge that this rule of thumb does not constitute such a model and can be executed without assessing the observer's perspective. This critique may be true – octopuses behaviour may be the result of a rule-of-thumb – but neither conclusion follows. How is such a rule of thumb a model? Implicit in moving under such a rule is the notion that 'if I can't see the threat, the threat can't see me'. The octopus does not have to cognitively represent this in awareness explicitly, or at all. This awareness is explicit in the movement. Even in humans, it is arguable that a basic level of awareness is comprised of our intuitive physics of the world. As we behave, we know what to expect. Hiding from a threat behind an object is useful whether or not the threat is an agent. How does following this rule-of-thumb necessitate assessing the observer's perspective? After all, sea urchins hold shells to block ultra-violet light damage to their tissues. The octopus's eyes are in the centre of the body. But the octopus is no apocryphal ostrich, hiding its head in the sand while its body is conspicuously in sight. The octopus moves so that the obstacle hides its body from the diver, and then brings only its eyes out of hiding to check on the threat.

Discussion

I have tried to show that octopuses recognize agency, have an awareness of themselves and others, and may thus form relationships with others in their environment. Octopuses in their behaviour discriminate among categories of others – that is, they behave differently toward predators than toward prey, and differently again to an agent like a SCUBA diver that is neither predator nor prey but with whom it is interesting to interact. Octopuses attend to the attention of observers, as shown by crawling under shelter in response to a predator, or waiting to make a sudden escape until a threating diver momentarily moves attention elsewhere. By its behaviour, an octopus also accounts for the observer's perspective, moving an object between itself and a diver before moving just its eyes back into view to keep tabs on the situation.

Previous authors have made arguments for primary animal consciousness using the argument of evolutionary homology (e.g. Edelman et al., 2005). However, as animals are more distantly related to humans, arguments from central nervous system organization become increasingly tenuous. The distributed organization of the octopus nervous system and the molluscan architecture of the octopus brain has undermined such arguments from anatomical homology.

Another line of argument has been from behavioural similarities (Mather, 2008), of which there are many. Like most vertebrates, octopuses appear to sleep (Brown et al., 2006; Meisel et al., 2011), a behaviour linked to a conflict between memory processing and behavioural dependence on visual information (e.g., Kavanau, 2006). Octopuses exhibit eye and arm lateralization (Byrne, 2004; Byrne et al., 2002, 2004, 2006). Octopuses also have complex and contingent behaviour that involves learning, spatial navigation, and decision making regarding food extraction, among other choices (e.g. see for review Mather & Scheel, 2014).

Other animals such as mammals are generally recognized as possessing at least primary consciousness (Edelman et al., 2005). Cognitive complexity, and with it some aspects of consciousness, have been linked to the need to understand another mind in social contexts (e.g. Borrego & Gaines, 2016). Thus, as social mammals, lions are a good choice to possess at least primary consciousness. During hunting, lions are aware of prey attention and vigilance, as well as pride members' behaviour and intentions. Similarly, octopuses are aware of categories of other agents around them (as predator, prey, curiosity, etc.), typical prey escape responses, possible predators and predator attention, and the relation of self to an observer. Behavioural similarities in these areas reveal similarity in awareness of agency in the world around them, and awareness of themselves in relation to other agents.

These are the minimal requirements on which to build relationships. It is yet unclear how common relationships with other agents are for octopuses in the wild, but we should expect the capability may be present. Anecdotes from both field work and captive animals suggest the importance of relationships with others in the lives of octopuses. This should not be surprising for octopuses or other animals with lifestyles involving complex interactions with the environment. Such interactions will necessitate discrimination of categories of individuals, the recognition of agency in others, and the need to place themselves and their complex, active bodies in relation to the intentions of threats, prey, potential mates and rivals, and possibly even friends that comprise the biologically relevant world around them.

Relationships are comprised of more than merely recognition and roles though. For example, what of the proximity-seeking and greetings of a dog to its owner (Prato-Previde et al., 2003)? What of learning to play a game with an owner, or encouraging the owner to go for a walk or get the food out? As yet, these aspects of expectation and interaction have not yet been studied among octopuses. The question of their occurrence, however, does not seem easily answered a priori or without investigation. Many animals recognize natural cycles (e.g. circadian rhythms), and exhibit behaviour (such as hunting) necessary to obtain reward (such as food). To the extent that this occurs, anticipation may also be a mechanism available to an animal with which to develop relationships. Some octopuses exhibit activity and rest cycles (Brown et al., 2006; Houck, 1982; Meisel et al., 2011), and speculatively forage for food. Giant Pacific octopuses (*Enteroctopus dofleini*) are also known to play (Mather & Anderson, 1999), although I am unaware of any report documenting interactive play between individuals.

Octopuses may recognize and remember other individuals, and possibly that allows them to form reciprocal relationships with others. If recognizing categories of others, and perhaps other individuals, are capabilities of octopuses, why has this not been more widely suspected? Boal's (2006) review found that cephalopod social abilities included recognition of species, sex and sexual receptivity, and perhaps dominance. Perhaps these are adequate to account for the behaviours I have described – for example, hierarchies may be accomplished without individual recognition if detectable features, such as size, are reliable predictors of dominance. Mate recognition may occur using a proxy such as proximity or place recognition of dens (Huffard, Caldwell & Boneka, 2008). The evidence that any octopus can recognize and remember other individuals is not yet strong, although there are hints (Morse et al., 2017; Tricarico et al., 2011). Opportunities to investigate individual recognition in octopuses are rare due to the solitary lives that most lead. This solitary lifestyle also suggests that even if octopuses have the ability to recognize other conspecific individuals, for many the ability will not often be used throughout their lives. Still, this ability may be within cognitive reach of octopuses. Another challenge has been that people are not good at recognizing individual octopuses from natural scars or markings. At the site in Jervis Bay, most octopuses are not individually distinguishable to us on the video. When an octopus enters the video frame, we are often unable to say with certainty whether that individual has been recorded before. In a few species or populations of octopuses, individuals have been recognizable by natural markings or scars (Caldwell et al., 2015; Huffard, Caldwell & Boneka, 2008; Huffard, Caldwell, DeLoach et al., 2008), but for most researchers, this is the exception.

Pet-keeping entails caretakers developing a relationship with the pet. In the cases of our most common pets, domesticated dogs, cats and birds, we expect the relationships to be reciprocated – the pet relates to the caretakers as well. As pet caretakers, we feel not only that the relationship is reciprocated, but also that the pet's well-being matters as an end in itself (e.g. Korsgaard, 2011).

What about the octopus? Octopuses are sometimes kept as pets (e.g., Dunlop & King, 2009; Mather et al., 2013). Are human-octopus relationships one-sided and therefore largely in the pet-keeper's imagination? Or are human-octopus relationships also octopus-human relationships? Mather et al. (2013) write that 'Octopuses watch us as much as we watch them.' Dunlop and King (2009) write that *Octopus bimaculoides* as pets 'will interact with their owners and enjoy physical contact such as being petted between and above the eyes'. *O. briareus* also 'is a friendly octopus, extending its long arms to touch the glass where your hand is'. These behaviours, watching and interacting with keepers, tracking their hands on the glass or movement around the room, responding to voices, approaching to be touched, are all commonly described by those who keep octopuses in any context. As described earlier, a negative experience can lead an octopus to avoid interaction or discourage it with a blast of water, and octopuses will thus develop different attitudes towards different people depending on their history together.

Many species of octopus are not well situated to live out their complete life-cycle in human society. Most coastal octopuses that make it into the aquarium trade tend eggs that hatch out into the plankton, and are challenging in captivity to raise until settlement. There are large-egged species with benthic juveniles that may be suitable candidates for aquaculture, but these species are not yet raised commercially. This remains a barrier to octopuses in regular domestic relationship with people, despite the ease with which they form relationships with those interested in interacting with them. Until this barrier is overcome, people and octopuses may form brief relationships when octopuses are grown up in captivity, or when divers visit octopuses in the sea.

Yet opportunities for people to have reciprocal relationships with octopuses seem to be increasing. Scheel et al. (2017) suggest that the social nature of at least some octopuses has been recognized increasingly over the past decade. There now are more people in the ocean as SCUBA divers, more octopuses in the ocean in general (e.g. Doubleday et al., 2016), and better technology in SCUBA diving, underwater photography and aquarium husbandry for observing octopuses. While most species of octopuses may not be domesticated due to the challenges of completing their life-cycle in human care, nevertheless we are increasingly getting to know them.

Acknowledgements

I thank Mark Faller and students on the Alaska Pacific University courses Philosophy of Science: Evolution and Philosophy of Science: Mind over many years, Peter Godfrey-Smith, and Stefan Linquist for discussions about some early versions of these ideas, some of which I also presented at Philosophy of Biology at Dolphin Beach for the 2013 meeting.

Funding

Field observations mentioned in this article were supported by Alaska Pacific University from donations by the Pollock Conservation Consortium. The research in Australia was conducted under permit number BDR16/00008 to P. Godfrey-Smith, D.S., and S. Chancellor to conduct research in Booderee National Park, and with approval of the Alaska Pacific University Institutional

Review Board. Findings and conclusions presented are the author's and do not necessarily reflect the views or positions of the supporting organizations.

Note

1 https://www.youtube.com/watch?v=_3x3xitCbCM.

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Author biography

David Scheel is a Professor of Marine Biology at Alaska Pacific University who specializes in theoretical, field and laboratory studies on the ecology and evolution of predator–prey relations and habitat use. His research since 1995 examines the biology of the giant Pacific octopus, and includes field studies in Prince William Sound, Alaska conducted in the intertidal, by SCUBA diving, and by submersible. He collaborates on behavioural studies of octopuses in Polynesia, Australia and the western Indian ocean.